Use work-energy to solve motion

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose work-energy when force and displacement are known.

SUCCESS CRITERIA - I can...

1. List all forces doing work.
2. Calculate the kinetic-energy change.
3. Check whether the result is physically possible.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “work” and “kinetic energy” to explain your model or result. Sentence starters are optional.

work: energy transferred by a force along a displacement

kinetic energy: energy associated with motion

final: at the end of the interval; subscript f

01 Bring back an earlier idea

A force parallel to motion does 15 J over 3 m.
Find its size. What work does the same force do with zero displacement?

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $W_{net} = \frac{1}{2}m(v_f^2 - v_i^2)$.
Constant mass. Use the displacement along each force component; stopping removes the initial kinetic energy.

Worked reasoning: A 2 kg cart starts from rest. Constant net force 4 N acts through 3 m, doing 12 J. Then $\frac{1}{2}mv^2=12$ gives $v=\sqrt{12}\approx3.46$ m/s.

Your turn: Sketch a force diagram and an energy bar diagram for this motion.
Identify where time is unnecessary.

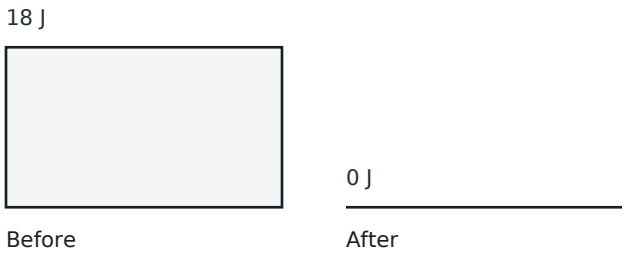
Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

A cart has 18 J kinetic energy and then stops. How much net work is done?
Show an energy bar before and after.

Sentence starter: I notice ... . This supports ... because ... .



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: ① ② ③

04 Solve and justify

A 2 kg cart moving at 6 m/s encounters a constant 3 N resistive force.
Find stopping distance and compare with the distance if initial speed doubles.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

A calculation gives negative final kinetic energy after a long braking distance. What should you check?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Choose a method

Initial momentum is 0.4 kg·m/s and measured final momentum is 0.38 kg·m/s.
Find percent deviation.
Explain why this is not itself a measured uncertainty.

Word help: momentum = mass times velocity, including direction

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: 1 2 3

07 Exit check

A 5 kg cart starts at rest. A 12 N pull and 2 N resistance act over 4 m.
Find final speed.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 4 kg cart at 5 m/s slows to 1 m/s over 6 m.
Find the constant net force component along its motion.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

Read down the left column, then down the right. Use the open space to draw, calculate and explain.